

Area, packing density, whitespace/deadspace

With a legal packing in hand, score the whitespace

Pseudocode

- Deadspace pseudocode is arithmetic after legality
- Sum module areas, subtract from outline area, and divide for density
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- On the golden packing area is twenty-three
- Do not celebrate density on illegal packs

Algorithm sketch — try these

```
INPUT: outline W×H, modules areas
OUTPUT: area_sum, deadspace, density
area_sum ←  $\sum w[m] \cdot h[m]$ 
deadspace ←  $W \cdot H - \text{area\_sum}$ 
density ←  $\text{area\_sum} / (W \cdot H)$ 
only report density on legal packs
GOLDEN: area=23; outline=80; dead=57
density=0.2875
```

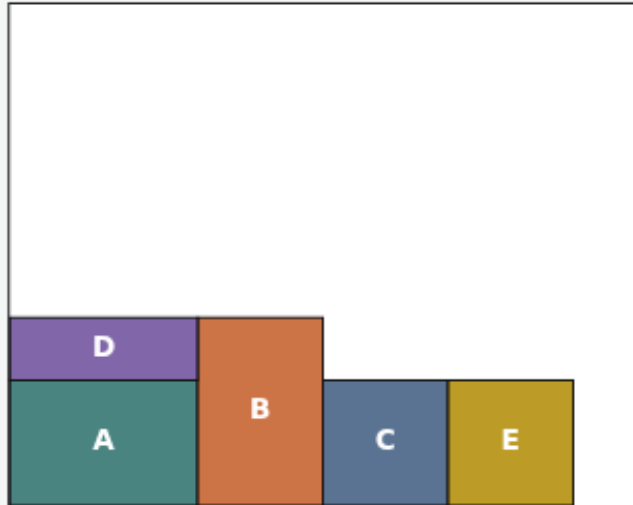
Module areas sum to 23

AREA & DEADSPACE

Module areas sum to 23

Step 1 / 5 · areas · module01-03-area-deadspace

10×8



Explanation

Add the five module areas: A is six, B is six, C is four, D is three, E is four. That is twenty-three units of silicon that must fit in the outline.

- $A \ 3 \times 2 = 6$
- $B \ 2 \times 3 = 6$
- $C \ 2 \times 2 = 4$
- $D \ 3 \times 1 = 3$
- $E \ 2 \times 2 = 4$

moduleAreaSum: 23

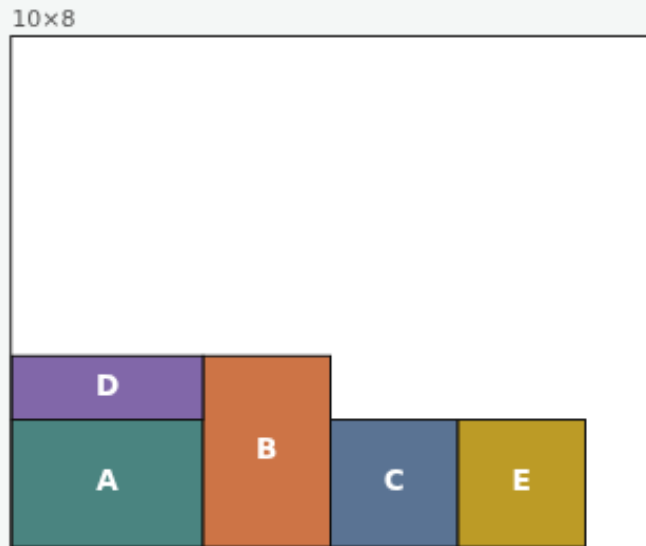
Module areas sum to 23

Outline area is 80

AREA & DEADSPACE

Outline area is 80

Step 2 / 5 · outline-area · module01-03-area-deadspace



Explanation

Ten times eight is eighty. That is the budget. Module area twenty-three leaves whitespace—deadspace—equal to fifty-seven.

- $\text{outlineArea} = W \times H = 80$
- $\text{deadspace} = 80 - 23 = 57$

```
outlineArea: 80  
deadspace: 57
```

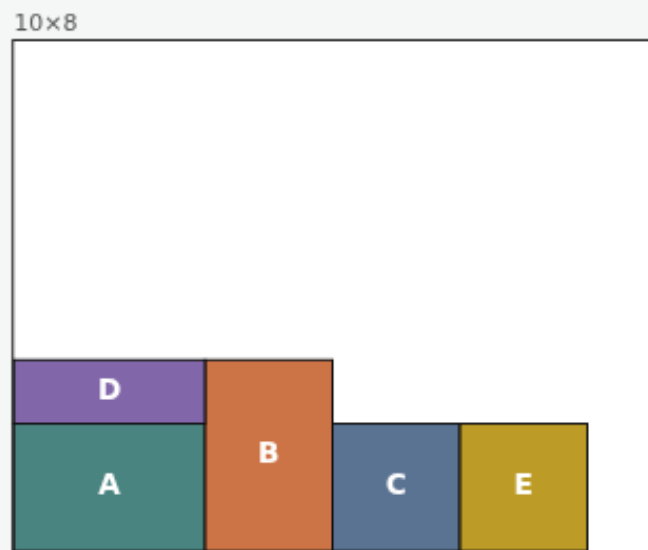
Outline area is 80

Density is 0.2875

AREA & DEADSPACE

Density is 0.2875

Step 3 / 5 · density · module01-03-area-deadspace



Explanation

Density is module area over outline area: twenty-three over eighty equals zero point two eight seven five. Whitespace fraction is zero point seven one two five.

- $\text{density} = 23/80 = 0.2875$
- $\text{whitespace} = 57/80 = 0.7125$
- Placement-independent for fixed sizes

density: 0.2875
whitespace: 0.7125

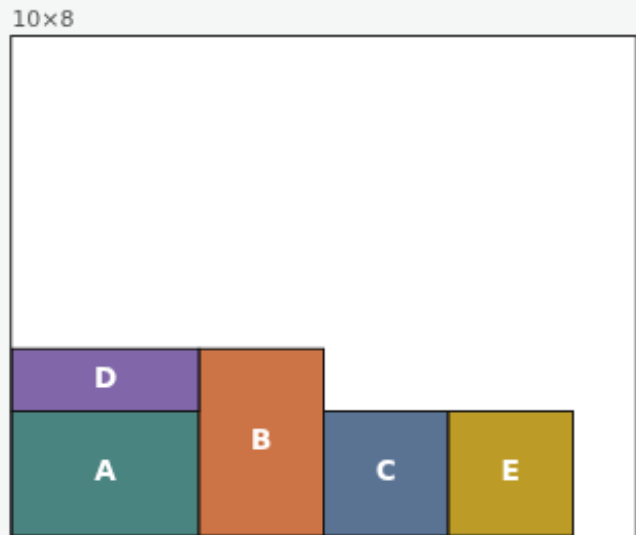
Density is 0.2875

Metrics need a legal pack

AREA & DEADSPACE

Metrics need a legal pack

Step 4 / 5 · legal-metrics · module01-03-area-deadspace



Explanation

On the golden legal packing these numbers are honest. On an overflow packing the area math still computes, but the floorplan is invalid—do not celebrate density on illegal layouts.

- Golden packing is legal
- Deadspace 57 is the teaching golden
- Illegal packs still report area math

```
legal: true  
deadspace: 57  
density: 0.2875
```

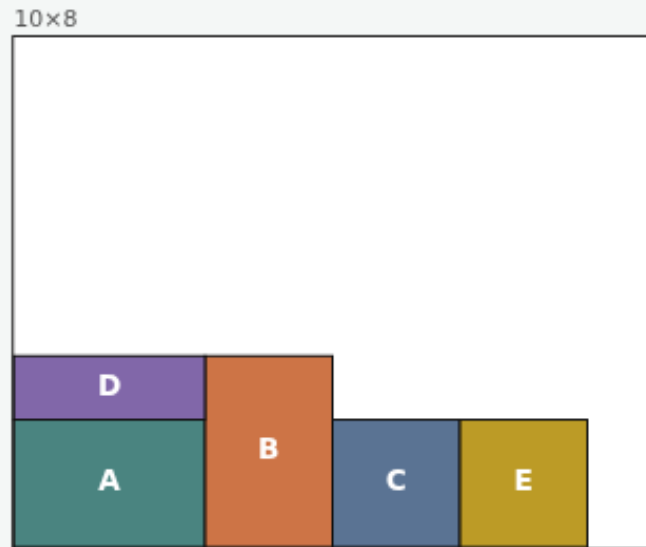
Metrics need a legal pack

Report area trio every time

AREA & DEADSPACE

Report area trio every time

Step 5 / 5 · takeaway · module01-03-area-deadspace



Explanation

Habit: print module area, outline area, deadspace, and density with every packing. Representations later change geometry; this trio stays the scoreboard.

- Always report 23 / 80 / 57 / 0.2875
- Density < 1 means whitespace remains
- Next: slicing polish that fits BB 9x3

golden: 23, 80, 57, 0.2875

Report area trio every time

Browser lab track

- Open area-deadspace, load the golden packing, and read the metrics panel

Implement track

- Implement `moduleAreaSum`, `outlineArea`, `deadspace`, and `density`
- Assert `deadspace` equals eighty minus twenty-three on the starter modules

Pitfalls

- Reporting density on an illegal pack

Your turn

- Print the area trio on every run
- Next lab builds a slicing polish whose bounding box is nine by three inside this outline